PH3244
Experiment - 3
FET Characteristics

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Synopsis

In this experiment we see the characterists of a FET transistor and calculate some properties of it.

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I. THEORY AND PROCEDURE

A. Theory[1]

AC Drain Resistance, r_D:
It is the AC resistance between drain and source terminals when JFET is operating in the pinch-off region. To calculate AC drain resistance calculate the slope of the drain characteristics in the pinch off region. r_D=change on V_{DS} change on I_D at V_{GS} constant or r_D= V_{DS} I_D at constant V_{GS} It has a very high value.

$$g_m = \frac{\text{change on } V_{DS}}{\text{change on } I_D} \text{at constant } V_{GS} \text{ or}$$

$$r_D = \frac{V_{DS}}{I_D} \text{at constant } V_{GS} \tag{1}$$

It has a very high value.

Transconductance, g_m:
To calculate transconductance determine slope of the transfer characteristics.

$$g_m = \frac{\text{change on } I_D}{\text{change on } V_{GS}} \quad \text{at constant } V_{GS} \tag{2}$$

Its unit is Siemens (S) / mho.
Amplification factor, μ :
It is given by

$$\mu = \frac{\text{change on } V_{DS}}{\text{change on } V_{DS}} \quad \text{at } I_D \text{ constant or}$$

$$\mu = \frac{V_{DS}}{V_{DS}} \quad \text{at constant } I_{DS}$$

$$\text{Or } \mu = g_m * r_d \tag{3}$$

DC drain resistance, R_{DS} :
It is also called the static or ohmic resistance of the channel. It is given by

$$R_{DS} = \frac{V_{DS}}{I_D} \tag{4}$$

B. Procedure

Refer to the manual [Cat (2026)] for a look at procedure, although the required pages will be attached at the back for sumbition along with the signed report.

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[1] Please refer to [Cat (2026)] for a full examination of theory, only the important results will be discussed here.

II. OBSERVATION

A. For constant V_{DS}

V_{GS} (V)	I_D (mA)
0.000	7.17
0.197	5.78
0.397	4.41
0.604	3.17
0.807	2.14
1.000	0.98
1.199	0.64

TABLE I: $V_{\text{DS}} = -2.65\text{V}$

V_{GS} (V)	I_D (mA)
0.000	8.58
0.202	6.86
0.402	5.23
0.684	3.94
0.799	2.76
0.999	1.75
1.203	0.44

TABLE II: $V_{\text{DS}} = -5\text{V}$

B. For constant V_{GS}

V_{DS} (V)	I_{GS} (mA)
-1.05	5.53
-2.01	7.15
-3.55	7.87
-4.04	8.27
-5.08	8.59
-6.02	8.79
-7.03	9.00
-8.08	9.16
-9.02	9.30
-10.01	9.42
-11.01	9.52
-12.07	9.59
-13.04	9.79

TABLE III: $V_{\text{GS}} = 0\text{V}$

V_{DS} (V)	I_{GS} (mA)
-1.09	0.08
-1.96	0.11
-3.02	0.15
-3.97	0.17
-5.05	0.20
-6.00	0.22
-7.04	0.25
-8.45	0.27
-8.98	0.30
-10.02	0.33

TABLE IV: $V_{\text{GS}} = 1.558\text{V}$

V_{DS} (V)	I_{GS} (mA)
-1.00	1.01
-2.05	1.30
-3.14	1.51
-4.15	1.65
-5.03	1.76
-6.03	1.88
-7.12	2.00
-8.01	2.09
-9.54	2.19
-10.00	2.27
-11.10	2.37
-12.04	2.46

TABLE V: $V_{\text{GS}} = 1\text{V}$

$V_{PO} = 2.3\text{V}$.
The least count is to be interpreted by the number of significant digits of that coloumn.
Data taken on 6 Feb 2026 by 20231289[2]

III. CALCULATIONS, ANALYSIS AND CHARACTERISTICS

A. Sources of Error

1. Systematic Errors

1. Heating of transistor
2. Resistance, Impidence and Capacitance of wiring and breadboard
3. Deviation from the actual value of resistor

[2] The data taken on 4 Feb 2026 is not written, for further details refer to Appendix - C.

2. Random Errors

1. Uncertanity in instruments

B. Error Propogation

To calculate the standard deviation from values we have measured with some least count, and the values we have applied digitally, we use the formula [Tewari (2016)]:

$$\sigma = \frac{\text{Least Count}}{\sqrt{12}} \tag{5}$$

and to calculate the standard deviation for values we have applied on analog instrument with some least count, we use the formula [Tewari (2016)]:

$$\sigma = \frac{\text{Least Count}}{\sqrt{24}} \tag{6}$$

Furthermore, we will calculate the uncertanity of r_D and g_m from the slope of their graph, to do that we will assume that the random error is much greater than the uncertainty calculatied form Eq - 5, 6. To calculate the μ we see from Eq - 3 and from [Taylor (1997)] that:

$$\Delta\mu = \mu\sqrt{\left(\frac{\Delta g_m}{g_m}\right)^2 + \left(\frac{\Delta r_d}{r_d}\right)^2} \tag{7}$$

C. Plots and Calculation[3]

First since we only want to see the characteristics of the transistors, we won't be taking any errors into consideration, We plot Table I-V and see:

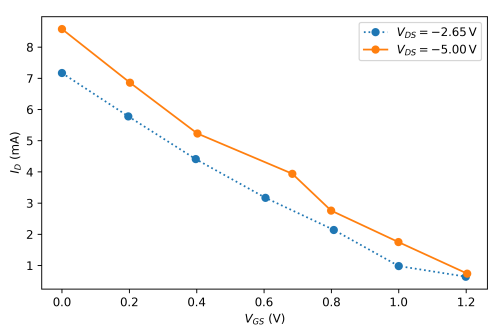

FIG. 1: I_d vs V_{GS} from Table - I, II

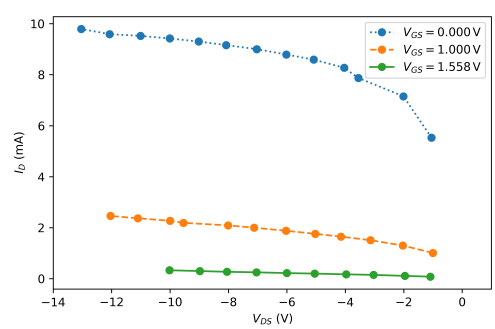

FIG. 2: I_d vs V_{DS} from Table - III, IV, V

Here we see that the characterists of Fig - 1 is linear whereas the Fig - 2 characterists suddenly increases form 0 to a high current and then stabalizes out.

[3] Refer to [Cat (2026)] for the code used to plot and determine the slope.

Now to calculate the r_D we shall take the region before $V_{DS} = -5V$ for linear best fit and for g_m we will do a best linear fit for the entire graph. Doing a best linear fit we get:

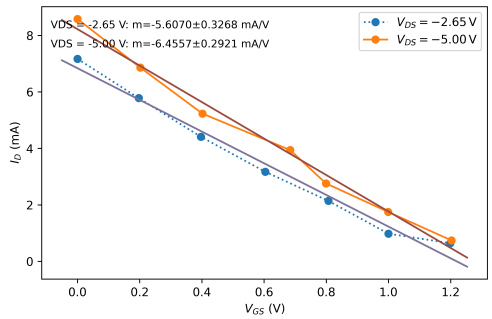

FIG. 3: Best linear fit for I_d vs V_{GS}

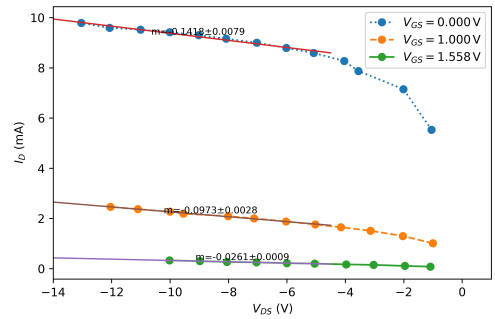

FIG. 4: Best linear fit for I_d vs V_{DS}

From this we get VGS = 0.000 V = -0.14185 ± 0.00786 mA/V
VGS = 1.558 V = -0.02615 ± 0.00087 mA/V
VGS = 1.000 V = -0.09733 ± 0.00276 mA/V
VDS = -2.65 V = -5.60698 ± 0.32677 mA/V
VDS = -5.00 V = -6.45574 ± 0.29215 mA/V

IV. RESULT AND CONCLUSION

$$\hat{p}_t = \underset{(1.958)}{0.900} + \underset{(0.017)}{0.997} p_{t-1}, \tag{8}$$

Cat (2026) Tewari (2016) Taylor (1997) sta (2026)

Appendix A: Checking the observed values with theoratical

here add the theo val adn comment if hter is any diff add tyhe 5th reference

Appendix B: Comments on how to improve the experiment

To calculate the DC drain resistance R_{DS}, there was not enough value taken in the interval of -1 - 0. Further measurements are needed to determine it.

To reduce the error due to heating, we should take the measuremeant as soon as possible to make the error as low as posssible.

Appendix C: Why the first dataset was discarded

The first dataset taken on 4 Feb 2026 is discarded and not written in the report as it was later found out that the FET given to us was faulty. A empty signed sheet of that date will

be attached for summision.

REFERENCES

L. M. Cat, Ph3244 3rd year sem 6 physics lab repo: https://github.com/laughingcatmeme/ph3244 (2026).
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